COMPANION FOR CHAPTER 4

OF

FUNDAMENTALS OF PROBABILITY

WITH STOCHASTIC PROCESSES

FOURTH EDITION

SAEED GHAHRAMANI

Western New England University
Springfield, Massachusetts, USA

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Chapter 4

Distribution Functions and Discrete Random Variables

4A ADDITIONAL EXAMPLES

Example 4a Flip a fair coin, if it lands heads let $X = 2$ and if it lands tails let X be a random point from the interval $(1, 3)$. Find the distribution function of X and use it to calculate $P(3/2 < X < 5/2)$.

Solution: The random variable X can only assume values of the interval $(1, 3)$, hence $F(t) = P(X \le t) = 0$ if $t \le 1$. To calculate $F(t)$ for other values of t, we use Theorem 3.3 of the book, the law of total probabilities. Let H and T stand for the events of heads and tails, respectively, when flipping the coin. Then

$$F(t) = P(X \le t) = P(X \le t \mid H)P(H) + P(X \le t \mid T)P(T).$$

Now from this relation, for $1 \le t < 2$ we get,

$$F(t) = 0 \times \frac{1}{2} + \frac{t-1}{3-1} \times \frac{1}{2} = \frac{t-1}{4},$$

and for $2 \le t < 3$ we get,

$$F(t) = 1 \times \frac{1}{2} + \frac{t-1}{3-1} \times \frac{1}{2} = \frac{t+1}{4}.$$

For $t \ge 3$, $F(t) = P(X \le t) = 1$. Hence

$$F(t) = \begin{cases} 0 & t < 1 \\ (t-1)/4 & 1 \le t < 2 \\ (t+1)/4 & 2 \le t < 3 \\ 1 & t \ge 3. \end{cases}$$

Observe that the magnitude of the jump at $t = 2$ is 1/2. Hence $P(X = 2) = 1/2$ as we expect. To calculate $P(3/2 < X < 5/2)$, note that

$$P(3/2 < X < 5/2) = P(X < 5/2) - P(X \le 3/2).$$

Now $P(X < 5/2) = P(X \le 5/2)$ since $P(X = 5/2) = 0$. Therefore,

$$\begin{aligned} P(3/2 < X < 5/2) &= P(X \le 5/2) - P(X \le 3/2) \\ &= F(5/2) - F(3/2) = \frac{(5/2)+1}{4} - \frac{(3/2)-1}{4} = \frac{3}{4} = 0.75. \quad \blacklozenge \end{aligned}$$

Example 4b In a certain country, where the resources are very limited, to test individuals for a specific disease, they divide them into groups of 15, take blood samples of all, mix them, and analyze the mixture. If the test is negative, none of the individuals in the group has the disease. If it is positive, then they must test each of the members of the group separately. If the probability that a person from this country has the disease, independently from others, is 0.07, and if for each group, it is chemically possible to mix their blood samples for the test, what is the expected number of tests for each group of 15 people?

Solution: In a random group, let X be the number of tests required to identify the individuals with the disease. Since the probability that a random person from this country does not have the disease is $1 - 0.07 = 0.93$,

$$X = \begin{cases} 1 & \text{with probability } (0.93)^{15} \\ 16 & \text{with probability } 1 - (0.93)^{15} \end{cases}$$

Therefore,

$$E(X) = 1 \cdot (0.93)^{15} + 16 \cdot \left[1 - (0.93)^{15}\right] \approx 10.95. \quad \blacklozenge$$

Example 4c Let X be a discrete random variable with distribution function

$$F(t) = \begin{cases} 0 & t < 1 \\ 1 - \left(\dfrac{4}{5}\right)^n & n \le t < n+1, \quad n = 1, 2, 3, \ldots. \end{cases}$$

Find p, the probability mass function of X and $E(X)$.

Solution: The set of possible values of X is $\{1, 2, 3, \ldots\}$, since it is at the points of this set that F has jumps. $P(X = i)$ is the magnitude of jump at i. We have

$$P(X = 1) = F(1) - F(1-) = \left(1 - \frac{4}{5}\right) - 0 = \frac{1}{5},$$

$$P(X = 2) = F(2) - F(2-) = \left[1 - \left(\frac{4}{5}\right)^2\right] - \left(1 - \frac{4}{5}\right) = \frac{4}{5} - \left(\frac{4}{5}\right)^2 = \frac{4}{5}\left(1 - \frac{4}{5}\right) = \frac{4}{5}\left(\frac{1}{5}\right).$$

In general, for $n \ge 1$,

$$\begin{aligned} P(X = n) = F(n) - F(n-) &= \left[1 - \left(\frac{4}{5}\right)^n\right] - \left[1 - \left(\frac{4}{5}\right)^{n-1}\right] = \left(\frac{4}{5}\right)^{n-1} - \left(\frac{4}{5}\right)^n \\ &= \left(\frac{4}{5}\right)^{n-1}\left(1 - \frac{4}{5}\right) = \left(\frac{4}{5}\right)^{n-1}\left(\frac{1}{5}\right). \end{aligned}$$

So the probability mass function of X is

$$P(n) = \Big(\frac{4}{5}\Big)^{n-1}\Big(\frac{1}{5}\Big), \qquad n = 1, 2, 3, \ldots.$$

By definition of expected value,

$$E(X) = \sum_{n=1}^{\infty} n\Big(\frac{4}{5}\Big)^{n-1}\Big(\frac{1}{5}\Big) = \frac{1/5}{4/5}\sum_{n=1}^{\infty} n\Big(\frac{4}{5}\Big)^{n}.$$

Since for $|r| < 1$, $\sum_{n=1}^{\infty} xr^x = r/(1-r)^2$, we have

$$E(X) = \frac{1}{4} \cdot \frac{\frac{4}{5}}{\Big(1 - \frac{4}{5}\Big)^2} = 5. \qquad \blacklozenge$$

Example 4d A box contains 5 red and 3 blue balls. The balls are identical except for their colors. Balls are drawn at random, one by one, and without replacement until a blue ball is drawn. Let X be the number of balls drawn. Find $E(X)$ and $\text{Var}(X)$.

Solution: The set of possible values for X is $\{1, 2, 3, 4, 5, 6\}$. The probabilities associated with these values are

$$P(X = 1) = \frac{3}{8};$$

$$P(X = 2) = \frac{5}{8} \cdot \frac{3}{7} = \frac{15}{56};$$

$$P(X = 3) = \frac{5}{8} \cdot \frac{4}{7} \cdot \frac{3}{6} = \frac{5}{28};$$

$$P(X = 4) = \frac{5}{8} \cdot \frac{4}{7} \cdot \frac{3}{6} \cdot \frac{3}{5} = \frac{3}{28};$$

$$P(X = 5) = \frac{5}{8} \cdot \frac{4}{7} \cdot \frac{3}{6} \cdot \frac{2}{5} \cdot \frac{3}{4} = \frac{3}{56};$$

$$P(X = 6) = \frac{5}{8} \cdot \frac{4}{7} \cdot \frac{3}{6} \cdot \frac{2}{5} \cdot \frac{1}{4} \cdot \frac{3}{3} = \frac{1}{56}.$$

Therefore,

$$E(X) = 1 \cdot \frac{3}{8} + 2 \cdot \frac{15}{56} + 3 \cdot \frac{5}{28} + 4 \cdot \frac{3}{28} + 5 \cdot \frac{3}{56} + 6 \cdot \frac{1}{56} = \frac{9}{4} = 2.25.$$

$$E(X^2) = 1 \cdot \frac{3}{8} + 4 \cdot \frac{15}{56} + 9 \cdot \frac{5}{28} + 16 \cdot \frac{3}{28} + 25 \cdot \frac{3}{56} + 36 \cdot \frac{1}{56} = \frac{27}{4};$$

$$\text{Var}(X) = E(X^2) - \big[E(X)\big]^2 = \frac{27}{4} - \Big(\frac{9}{4}\Big)^2 = \frac{27}{16} \approx 1.69.$$

Alternative Solution: To calculate the probability mass function of X, $p(i) = P(X = i)$, $i = 1, 2, \ldots, 6$, let A_i be the event of $i - 1$ red balls in i draws and B_i be the event that in i draws, the last draw is a blue ball. Then

$$p_i = P(X = i) = P(A_iB_i) = P(A_i)P(B_i \mid A_i) = \frac{\binom{5}{i-1}\binom{3}{1}}{\binom{8}{i}} \cdot \frac{1}{i}.$$

Therefore,

$$E(X) = \sum_{i=1}^{6} i \cdot \frac{\binom{5}{i-1}\binom{3}{1}}{\binom{8}{i}} \cdot \frac{1}{i} = \frac{9}{4} = 2.25.$$

$$E(X^2) = \sum_{i=1}^{6} i^2 \cdot \frac{\binom{5}{i-1}\binom{3}{1}}{\binom{8}{i}} \cdot \frac{1}{i} = \frac{27}{4};$$

$$\text{Var}(X) = E(X^2) - \big[E(X)\big]^2 = \frac{27}{4} - \left(\frac{9}{4}\right)^2 = \frac{27}{16} \approx 1.69. \quad \blacklozenge$$

Example 4e Let X be a discrete random variable with the set of possible values $\{1, 2, 3 \ldots\}$ and finite expected value. Show that

$$E(X) = \sum_{n=1}^{\infty} P(X \geq n).$$

Solution: By the definition of expected value,

$$\begin{aligned} E(X) &= \sum_{i=1}^{\infty} iP(X = i) = \sum_{i=1}^{\infty} \sum_{n=1}^{i} P(X = i) \\ &= \sum_{n=1}^{\infty} \sum_{i=n}^{\infty} P(X = i) = \sum_{n=1}^{\infty} P(X \geq n). \quad \blacklozenge \end{aligned}$$

Example 4f Friday the 13th, How Often Do They Occur? Not only superstitious people, but also non-superstitious mathematicians have shown interest in the number of Friday the 13th's that appear in a given year. Let X be the number of Friday the 13th's in a randomly chosen year.

(a) Find the probability mass function of the random variable X.

(b) What is the probability that the 13th of a randomly selected month is a Friday?

(c) What is the probability that a randomly selected year has two consecutive months with their 13th's falling on Friday?

Solution: **(a)** The following table determines the months of any given year that have their 13th on a Friday.

MONTHS WITH A FRIDAY THE 13TH

Day Number	Day	Non-Leap Year	Leap Year
0	Sunday	January, October	January, April, July
1	Monday	April, July	September, December
2	Tuesday	September, December	June
3	Wednesday	June	March, November
4	Thursday	February, March, November	February, August
5	Friday	August	May
6	Saturday	May	October

This table is given by Ritter et al. in their paper "An Aid to the Superstitious," which was published in vol. 70, no. 5 issue of the *Mathematics Teacher* in 1977. To use the table above, all we need to know is the first day of the year. For example, if the first day of a non-leap year is a Thursday, then that year has three months with their 13th's falling on Friday, namely, February, March, and November. To find the first day of a given year, Y, let

$$f(Y) = Y + [(Y-1)/4] - [(Y-1)/100] + [(Y-1)/400], \tag{1}$$

where the symbol $[x]$ denotes the greatest integer less than or equal to x. Then find the D, the number that satisfies

$$f(Y) \equiv D \pmod 7, \tag{2}$$

where by $a \equiv b \pmod n$ we mean that when a is divided by n, the remainder is b. If D is 0, the year begins on a Sunday. If it is 1, the year begins on a Monday, and so on. As an example, consider the year $Y = 1992$. From (1) we have,

$$\begin{aligned} f(1992) &= 1992 + [1991/4] - [1991/100] + [1991/400] \\ &= 1992 + 497 - 19 + 4 = 2474. \end{aligned}$$

Now dividing 2474 by 7, the remainder obtained is 3. Thus $f(1992) \equiv 3 \pmod 7$, and hence the first day of 1992 is a Wednesday. But 1992 is a leap year, so from the table above we have that this year has two months with their 13th's falling on Friday, namely, March and November. The underlying reason for (1) is that the Gregorian Calendar has a leap year every four years except century years that are not divisible by 400. Thus all non-century years divisible by 4 and all century years divisible by 400 are leap years. For example, 1988, 2000, 2060, 2296, and 2800 are leap years while 1300, 1977, 1998, 2067, 2100, and 2700 are not.

Now the set of possible values of X, the number of the Friday the 13th's in a randomly selected year, is $\{1, 2, 3\}$. To find $P(X = i)$, $i \in \{1, 2, 3\}$, we will use the knowledge that the Gregorian Calendar is periodic with period of 400. By this knowledge, for any year Y, let Z be one of the years Y through $Y + 399$. Then whatever day of the week on which

January 1st of Z falls, the January 1st of $Z+400$ will fall on the same day of week. Thus the respective numbers of Friday the 13th's in the years Y through $Y+399$ are the same as the numbers of Friday the 13th's in the years $Y+400$ through $Y+799$. Let us restrict ourselves to the period from the year 2000 to the year 2399. For $0 \le i \le 6$, let $|L_i|$ and $|NL_i|$ be the respective numbers of leap and non-leap years in this period with their January 1st a number i day. Then after using (1) and (2) to determine the days of the week that the January 1st's of the years 2000 to 2399 fall on, we obtain

$$\begin{aligned}
|NL_0| &= |NL_1| = |NL_3| = |NL_5| = |NL_6| = 43,\\
|NL_2| &= |NL_4| = 44,\\
|L_0| &= |L_5| = 15,\\
|L_1| &= |L_4| = |L_6| = 13,\\
|L_2| &= |L_3| = 14.
\end{aligned}$$

Now, it must be clear that, for $i = 1, 2, 3$, the fraction of all years with exactly i Friday the 13th's is equal to the fraction of the years in one period with exactly i Friday the 13th's. So

$$\begin{aligned}
P(X=1) &= \frac{|NL_3| + |NL_5| + |NL_6| + |L_2| + |L_5| + |L_6|}{400}\\
&= \frac{43+43+43+14+15+13}{400} = \frac{171}{400}.
\end{aligned}$$

Similar arguments show that, $P(X=2) = 170/400$ and $P(X=3) = 59/400$.

(b) Using the same method, we have that the probability that the 13th of a randomly selected month will be Friday is $688/(400 \times 12) = 43/300$, where 688 is obtained from the following formula:

$$\begin{gathered}
2|NL_0| + 2|NL_1| + 2|NL_2| + |NL_3| + 3|NL_4| + |NL_5| + |NL_6| + \\
3|L_0| + 2|L_1| + |L_2| + 2|L_3| + 2|L_4| + |L_5| + |L_6| = 688.
\end{gathered}$$

(c) The probability that a randomly selected year has two consecutive months with their 13th's falling on Friday is $|NL_4|/400 = 44/400 = 11/100$. ♦

The following, a *truly challenging problem*, is an example of a game in which despite a low probability of winning, the expected length of the play is high.

Example 4g (The Clock Solitaire) An ordinary deck of 52 cards is well shuffled and dealt face down into 13 equal piles. The first 12 piles are arranged in a circle like the numbers on the face of a clock. The 13th pile is placed at the center of the circle. Play begins by turning over the bottom card in the center pile. If this card is a king, it is placed face up on the top of the center pile, and a new card is drawn from the bottom of this pile. If the card drawn is not a king, then (counting the jack as 11 and the queen as 12) it is placed face up on the pile located in the hour position corresponding to the number of the card. Whichever pile the card drawn is placed on, a new card is drawn from the bottom of that pile. This card is placed face up on the pile indicated (either the hour position or the center depending on whether the card is or is not a king) and the play is repeated. The game ends when the 4th king is placed on the center pile. If that occurs on the last remaining card, the player wins. The number of cards turned over until the 4th king appears determines the length of the game. Therefore, the player wins if the length of the game is 52.

(a) Find $p(j)$, the probability that the length of the game is j. That is, the 4th king will appear on the jth card.

(b) Find the probability that the player wins.

(c) Find the expected length of the game.

Solution:

(a) The sample space has $52!$ elements because when the cards are dealt face down, any ordering of the cards is a possibility. To find $p(j)$, the probability that the 4th king will appear on the jth card, we claim that in $\binom{4}{1} \cdot {}_{(j-1)}P_3 \cdot 48!$ ways the 4th king will appear on the jth card, and the remaining 3 kings earlier. To see this, note that we have $\binom{4}{1}$ combinations for the king that appears on the jth card, and ${}_{(j-1)}P_3$ different permutations for the remaining 3 kings that appear earlier. The last term $48!$, is for the remaining 48 cards that can appear in any order in the remaining 48 positions. Therefore,

$$p(j) = \frac{\binom{4}{1} \cdot {}_{(j-1)}P_3 \cdot 48!}{52!} = \frac{\binom{j-1}{3}}{\frac{52!}{4!\,48!}} = \frac{\binom{j-1}{3}}{\binom{52}{4}}.$$

(b) The probability that the player wins is $p(52) = \binom{51}{3} \Big/ \binom{52}{4} = 1/13.$

(c) To find

$$E = \sum_{j=4}^{52} j p(j) = \frac{1}{\binom{52}{4}} \sum_{j=4}^{52} j \binom{j-1}{3},$$

the expected length of the game, we use a technique introduced by Jenkyns and Muller in *Mathematics Magazine*, 54, (1981), page 203. We have the following relation which can be readily checked.

$$j\binom{j-1}{3} = \frac{4}{5}\left[(j+1)\binom{j}{4} - j\binom{j-1}{4}\right], \quad j \geq 5.$$

This gives

$$\begin{aligned}\sum_{j=5}^{52} j\binom{j-1}{3} &= \frac{4}{5}\left[\sum_{j=5}^{52}(j+1)\binom{j}{4} - \sum_{j=5}^{52} j\binom{j-1}{4}\right]\\ &= \frac{4}{5}\left[53\binom{52}{4} - 5\binom{4}{4}\right] = 11,478,736,\end{aligned}$$

where the next-to-the-last equality follows because terms cancel out in pairs. Thus

$$\begin{aligned}E &= \frac{1}{\binom{52}{4}}\sum_{j=4}^{52} j\binom{j-1}{3} = \frac{1}{\binom{52}{4}}\left[4 + \sum_{j=5}^{52} j\binom{j-1}{3}\right]\\ &= \frac{1}{\binom{52}{4}}(4 + 11,478,736) = 42.4.\end{aligned}$$

As Jenkyns and Muller have noted, “This relatively high expectation value is what makes the game interesting. However, the low probability of winning makes it frustrating!” ♦

4B AN INVESTMENT APPLICATION

In business, **financial assets** are instruments of trade or commerce valued only in terms of the national medium of exchange (i.e., money) having no other intrinsic value. These instruments can be freely bought and sold. They are, in general, divided into three categories. Those guaranteed with a fixed income or income based on a specific formula are **fixed-income securities.** An example would be a bond. Those that depend on a firm's success through the purchase of shares are called **equity.** An example would be common stock. If the financial value of an asset is derived from other assets or depends on the values of other assets, then it is called a **derivative security.** For example, the interest rate of an adjustable-rate mortgage depends on a combination of factors concerning various other interest rates that determine an interest rate index, which in turn determines the periodical adjustments to the mortgage rate.

For the purposes of simplicity we are going to assume that the fixed-income securities are in the form of zero-coupon bonds, meaning they pay no annual interest but provide a return to the investor based on the capital gain over the purchase price. Similarly, we will assume that the instruments based on the firm's success represent businesses that pay no annual dividends but instead promise a return to the investor based on the growth of the company and thus the value of the stock.

Let X be the amount paid to purchase an asset, and let Y be the amount received from the sale of the same asset. Putting fixed-income securities aside, the ratio Y/X is a random variable called the **total return** and is denoted by R. Obviously, $Y = RX$. The ratio $r = (Y - X)/X$ is a random variable called the **rate of return**. If we purchase an asset for \$100 and sell it for \$120, the total return is 1.2, whereas the rate of return is 0.2. The latter quantity shows that the value of the asset was increased 20%, whereas the former shows that its value reached 120% of its original price. Clearly, $r = (Y/X) - 1 = R - 1$, or $R = 1 + r$.

Every investor has a collection of financial assets, which can be kept, added to, or sold. This collection of financial assets forms the investor's **portfolio**. Let X be the total investment. Suppose that the portfolio of the investor consists of a total of n financial assets. Furthermore, let w_i be the *fraction* of investment in the ith financial asset. Then $X_i = w_i X$ is the amount invested in the ith financial asset, and w_i is called the **weight** of asset i. Clearly,

$$\sum_{i=1}^{n} X_i = \sum_{i=1}^{n} w_i X = X$$

implies that $\sum_i^n w_i = 1$. If R_i is the total return of financial asset i, then asset i sells for $R_i w_i X$, and R, the total return is obtained from

$$R = \frac{\displaystyle\sum_{i=1}^{n} R_i w_i X}{X} = \sum_{i=1}^{n} w_i R_i. \tag{3}$$

Similarly, r, the rate of return of the portfolio is

$$r = \frac{\sum_{i=1}^{n} R_i w_i X - X}{X} = \sum_{i=1}^{n} R_i w_i X - 1 = \sum_{i=1}^{n} R_i w_i - \sum_{i=1}^{n} w_i$$
$$= \sum_{i=1}^{n} w_i (R_i - 1) = \sum_{i=1}^{n} w_i r_i. \quad (4)$$

We have made the following important observations:

> The total return of the portfolio is the weighted sum of the total returns from each financial asst of the portfolio.
>
> The rate of return of the portfolio is the weighted sum of the rates from the assets of the portfolio.

Putting fixed-income securities aside, for each i, R_i and r_i are random variables, and we have the following formulas for the expected values of R and r:

$$E(R) = \sum_{i=1}^{n} w_i E(R_i), \quad (5)$$

$$E(r) = \sum_{i=1}^{n} w_i E(r_i). \quad (6)$$

These formulas are simple results of Theorem 10.1 proved in Chapter 10 of the book. That theorem is a generalization of the corollary following Theorem 4.2 of the book. ⧫

4C QUANTILES AND THE MEDIAN

Let X be a random variable with the distribution function F, then Q_p is said to be a *quantile of order* p *of* X if

$$P(X \leq Q_p) \geq p \quad \text{and} \quad P(X \geq Q_p) \geq 1 - p.$$

This is equivalent to

$$P(X < Q_p) \leq p \leq P(X \leq Q_p)$$

or

$$F(Q_p-) \leq p \leq F(Q_p).$$

Therefore Q_p is a value at which either F is p or F jumps from under p to over p. If X is a continuous random variable, and f is its probability density function, then the area under the graph of f to the left of Q_p is p and the area to the right of Q_p is $1 - p$.

Whether X is continuous or discrete, the quantity $Q_{.5}$ is called the *median*, $Q_{.25}$, the *first quartile* and $Q_{.75}$ the *third quartile* of X. The difference $Q_{.75} - Q_{.25}$ is called the *interquartile range*. If X is continuous, then since half of the area under the graph of f lies to the left of the median and the other half to its right, the median is often used as a *center* of the distribution.

Quantiles are important measures of location and are very useful both in the analysis of data and random variables. To demonstrate how they are calculated, we will give two examples.

Example 4h. From the Webster's New World Dictionary we selected 30 words at random and found that 6 of them had 1 vowel, 6 of them 2, 10 of them 3, 6 of them 4, and 2 of them 5 vowels. Let X be the number of vowels in a random word from these 30. Then

$$\begin{aligned}
&P(X = 1) = P(X = 2) = P(X = 4) = 6/30,\\
&P(X = 3) = 10/30,\\
&P(X = 5) = 2/30.
\end{aligned}$$

Therefore F, the distribution function of X is given by

$$F(x) = \begin{cases} 0 & x < 1 \\ 6/30 & 1 \leq x < 2 \\ 12/30 & 2 \leq x < 3 \\ 22/30 & 3 \leq x < 4 \\ 28/30 & 4 \leq x < 5 \\ 1 & x \geq 5. \end{cases}$$

Now $Q_{.5}$, the median of X is a value at which either F is .5 or F jumps from under .5 to over .5. Therefore $Q_{.5} = 3$. Note that $P(X < 3) \leq .5 \leq P(X \leq 3)$ is valid only at 3, so the median in this case is unique. However, $Q_{.2}$, for instance, is not unique. $Q_{.2}$ is a value at which either F is .2 or F jumps from under .2 to over .2. All of the points of the interval $[1, 2)$ have this property. Therefore each of them can be taken as $Q_{.2}$. ♦

Example 4i. Let X be the amount of time (in minutes) that a randomly selected customer waits to be served in a restaurant. Suppose that the distribution of X is given by $F(t) = 1-e^{-t/15}$. To determine $Q_{.5}$, the median of F, note that since $F(Q_{.5}-) \leq .5 \leq F(Q_{.5})$ and F is continuous, we have that $F(Q_{.5}) = .5$. This gives $1 - e^{-Q_{.5}/15} = 1/2$. or $e^{-Q_{.5}/15} = 1/2$. The solution to this equation is $Q_{.5} = 15 \ln 2 \approx 10.40$. Therefore the probability that a customer has to wait more than 10.40 minutes is .5 and less than 10.40 is also .5. In other words, roughly, half of the customers wait more than 10.40 minutes and half of them wait less. ♦